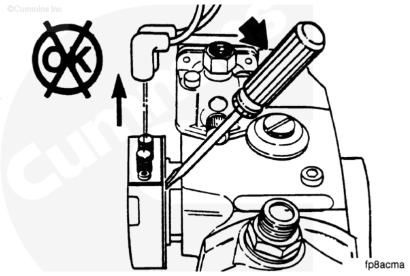
Trainings

Remove

⚠ CAUTION ⚠

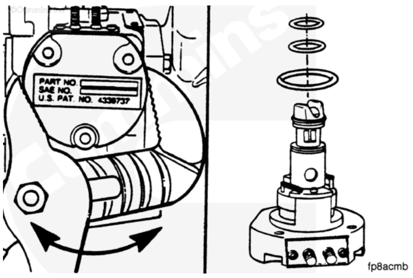
Do not pry the actuator from the housing. This can damage the actuator shaft and cause sticking.

Remove the actuator wires and capscrews.



Twist the actuator and pull it from the housing.

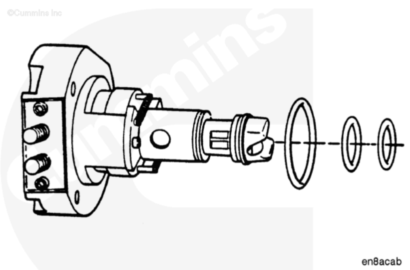
Remove the three o-rings from the actuator.



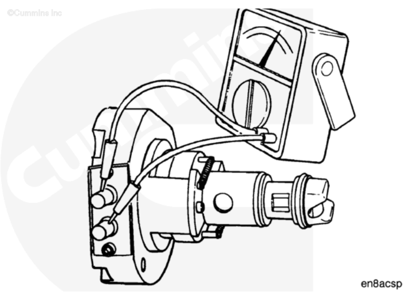
Install

Install a new o-ring on the 50 mm [2.0 in] diameter of the actuator.

Install two new o-rings on the actuator barrel.



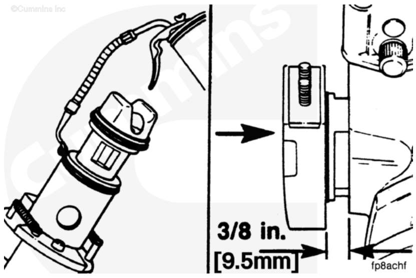
Apply the actuator rated battery voltage across the two terminals on the actuator to test the solenoid and to observe actuator operation. The actuator will make a loud click when the actuator shaft hits the internal stop. Removing the voltage from the actuator terminals will allow the force of the springs to return the actuator shaft to its original position. A click **must** be heard when the voltage is removed.



Note : The electronic fuel control (EFC) housing does not require the EFC plug in the bottom of the EFC housing bore.

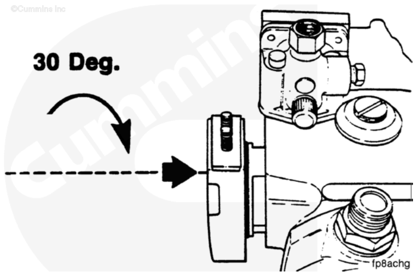
Lubricate the two barrel o-rings with clean engine oil.

Insert the actuator in the fuel pump housing. The actuator flange will be approximately 9.5 mm [0.37 in] from the fuel pump housing.

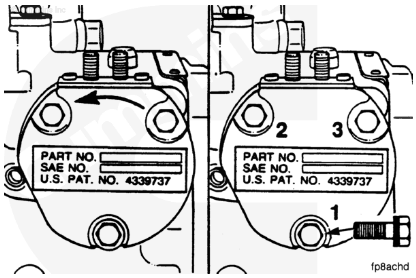


Use the palm of the hand to firmly press and rotate the actuator approximately 30 degrees until the actuator flange contacts the fuel pump housing.

Rotate the actuator until the mounting holes are aligned.

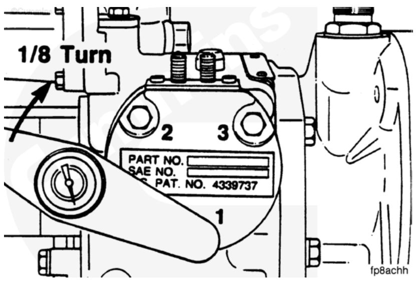


Install the three 1/4-20 x 1 1/4-inch hex head capscrews. These capscrews have captive spring washers and do **not** require lockwashers. Tighten the capscrews until they are finger-tight.



The actuator capscrews **must** be tightened in the following sequence:

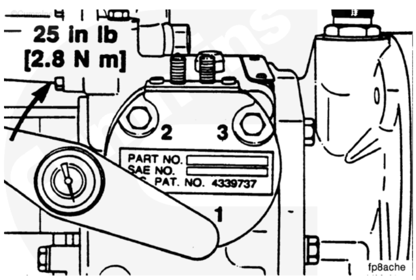
1. Tighten the mounting capscrews 1/8 of a turn, in the sequence shown in the figure, until they are seated.



2. Tighten the capscrews in sequence.

Torque Value:

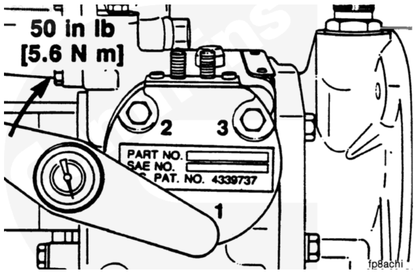
1. 2.8 n•m [25 in-lb]



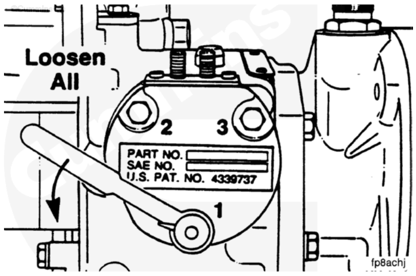
3. Tighten the capscrews in sequence.

Torque Value:

1. 5.6 n•m [50 in-lb]



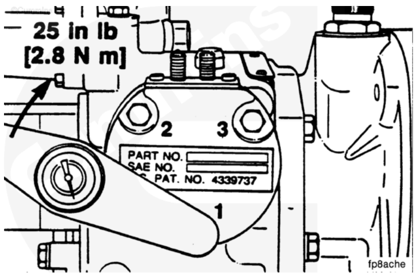
4. Loosen all three capscrews completely.



5. Tighten the capscrews again in sequence.

Torque Value:

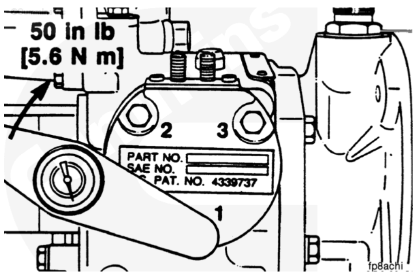
1. 2.8 n•m [25 in-lb]



6. Tighten the capscrews again in sequence.

Torque Value:

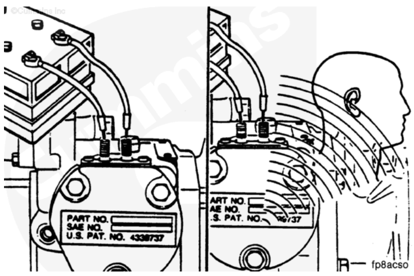
1. 5.6 n•m [50 in-lb]



7. This procedure will make sure that the actuator is properly installed and **not** binding.

⚠ CAUTION ⚠

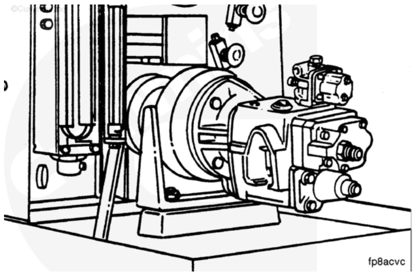
This test will only verify that the actuator will go from the full open to the full closed position. A slight binding of the actuator shaft can cause a governor stability complaint. This test may not detect a slight binding.



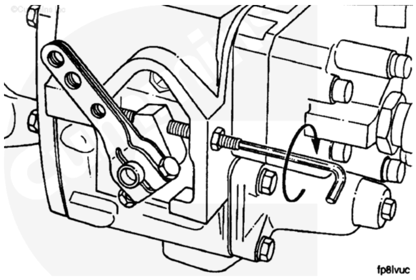
A final check is to apply and remove battery voltage across the two actuator terminals. The operation of the actuator **must** have a sound similar to what it did before installing in the fuel pump housing. If the actuator does **not** click, as if it is **not** operating, or operating slower than before, loosen all of the capscrews and tighten them again as described in the previous procedure.

The fuel pump can now be calibrated (reference the fuel pump calibration manuals or the monthly Cumulative Supplement Update).

Apply the actuator-rated battery voltage to the normally closed actuator when the fuel pump is calibrated.



The throttle shaft **must** be locked in the full open position. After the calibration, the fuel pump can be mounted on the engine.



Last Modified: 15-Oct-2019